GE5251 ENVIRONMENTAL SCIENCES

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ASSIGNMENT

1) What is timber extraction? Elaborate its uses and consequences.

Timber/wood extraction is a pheromenon increasing day by day. Due to population growth and the lack of alternative fuels, people living nearby forest areas use wood as their fuel. Hence, timber extraction is a very important process in these regions.

USES

→ Timber is used as raw materials for various wood-based industries

- ★ pulp and paper
- ★ composite wood
- ★ furniture

→ Timber is used for various developmental activities

- ★ railways
- ★ boats
- ★ road construction

CONSEQUENCES

→ Large scale timber extraction causes deforestation

→ Leads to soil erosion
loss of fertility
landslides
loss of biodiversity

→ Leads to loss of tribal culture and extinction of tribal people

→ Reduces thickness of the forest.

In India, industries consume about 28 million cu. mts/year of wood. However, annual forest growth is only about 12 million cu.mts/year.

2) Justify: 'Equitable use of resources helps in achieving sustainable development.'

Sustainable development is the development of a healthy environment without damaging the natural resources. In other words, all the natural resources must be used in such a way that it must be available for future generations.

So, unsustainable development means the degradation of the environment due to overutilisation and overexploitation of the natural resources.

In developed countries - which consists of only 22% of the world's population - 88% of the world's natural resources are being used. Their income contributes to nearly 85% of total global income. However, as mentioned before, they consist of less than 1/4 of the world's population. So in less developed countries, which comprise of 78% of the world's population, there is humongous lack of resources and income. The scarcity leads to unsustainability.

Rich countries consume more resources with more income, whereas in poor countries, there is overpopulation, and consumption of resources is too low.

In order to achieve sustainability, it is essential to achieve a more balanced and equitable distribution of land resources. Rich countries should lower down their consumption levels, and the minimum needs of the poor should be fulfilled by providing them resources.

Thus, a more balanced and equitable use of resources will reduce the differences between the more developed and less developed countries, and will lead to sustainable development.

3) Discuss the uses and ill effects of dam construction.

Dams are massive artificial structures built across the river to create a reservoir inorder to store water for many beneficial purposes.

→ The reservior can be used for irrigation purposes, water sports or even as other forms of pleasurable activities.

→ Dams can be used to generate electricity in ~~hydropower~~ hydroelectric power stations, which is a clean form of energy that doesn't contribute to pollution.

→ Dams help to control devastating floods, that prevent the loss of life and property.

However, dams are also responsible for the destruction of vast areas of forest and displacement of local people.

→ Thousands of hectares of forest have been cleared for executing river valley projects, hydroelectric projects, and the like.

→ Dam construction leads to the killing of wild animals and destroying aquatic life.

→ Displacement and cultural change due to dam construction affects tribal people both mentally and physically. They received no compensation, and will be affected by many diseases.